

PROJECT OVERVIEW

SCOREX • Version 1.0

01. Executive Summary

ScoreX is an Android cricket scoring application built to manage a match from initial setup through live delivery entry, innings management, statistics, scorecard review, graphs, local match history and PDF scorecard generation.

The central design idea is delivery-level state. A single delivery can affect the team score, batsman runs, balls, bowler statistics, legal-ball count, strike, partnership and wicket state. ScoreX keeps these related changes coordinated through its scoring/model layer rather than treating the score as a single text value.

The current project is a local Android application. It does not contain a traditional relational database or cloud backend in the supplied source. Match history uses Java serialization, settings use SharedPreferences, and generated PDFs are kept locally.

Project identity

Item	Current project value
Application	ScoreX Cricket Scorer
Application ID	com.apocalypes.scorex
Version	1.0
Language	Java
Java compatibility	11
Minimum SDK	24
Compile / Target SDK	37
Storage approach	Local files + SharedPreferences
Traditional database	Not used

02. Problem Statement, Objectives & Scope

Problem Statement

Manual cricket scoring requires continuous calculations and synchronized tracking. When a wide, no-ball, wicket, partnership change or run-out occurs, several statistics may need to change at the same time. A small mistake can therefore propagate into the scorecard and final result.

ScoreX addresses this by recording delivery events and maintaining structured information for players, bowlers, overs, partnerships and innings.

Objectives

- Provide an easy-to-use Android interface for live cricket scoring.
- Reduce repetitive manual calculations during matches.
- Maintain consistent batting, bowling, partnership and innings information.
- Provide a structured scorecard for match review.
- Provide graphs for visual scoring analysis.
- Store completed matches locally for later review.
- Generate shareable PDF scorecards.

Scope

The current scope covers local match scoring and review. It includes match setup, live scoring, special deliveries, wickets, innings transitions, result calculation, scorecard, graphs, history, settings and PDF export. It does not claim cloud synchronization, online accounts or a remote database.

CORE FEATURE	WHAT IT DOES
	Teams, overs, toss winner and decision.
Advanced Settings	Players-per-team and special-delivery behavior.
	Striker, non-striker and opening bowler.
Live Scoring	Normal runs and match controls.
	Wide, No Ball, Bye and Leg Bye.
Wickets	Dismissal and wicket state.
	Completed runs are retained on Run Out.
Undo	Reverses the most recent delivery.
	Current/completed partnership information.
Bowling	Overs, runs, wickets and special deliveries.
	Delivery and over-level match information.
Innings	First and second innings.
	Final result after completion.
Scorecard	Batting, extras, bowling, FOW and partnerships.

CORE FEATURE	WHAT IT DOES
	Runs-per-over, rate and innings comparison.
Match History	Local save/load/delete.
	Formatted scorecard documents.
Settings	Preferences, About and static Google link.

SYSTEM DESIGN

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03. Technology & System Requirements

Technology Stack

Layer	Technology / value
Mobile platform	Android
Programming language	Java
Java compatibility	11
Minimum SDK	24
Compile / Target SDK	37
UI	AndroidX + Material components
Local settings	Android SharedPreferences
Match persistence	Java serialization (.ser)
PDF output	Local scorecard PDF files

Hardware Requirements

- Android smartphone or compatible emulator.
- Android API 24 or later.
- Sufficient local storage for match history and generated PDFs.
- Touch input for the scoring interface.

Software Requirements

- Android Studio and a compatible Android SDK.
- Java 11-compatible build environment.
- Gradle configuration supplied with the project.

Design Principle

The application is designed to work for ordinary scoring without a network service. This makes the core scorer suitable for situations such as grounds where connectivity may be unreliable.

Architecture at a Glance

USER	PRESENTATION	SCORING CORE	DATA / OUTPUT
Scorer enters match events	Activities & UI Live Match • Scorecard • Graphs	MatchManager Models • Rules • Undo	History • Settings PDF • Local Files

04. System Architecture & Modules

Architectural Flow

The supplied project separates screen-level responsibilities from the scoring/model layer. Activities provide user interaction, while MatchManager and the model classes represent the state of the match. Persistence and PDF output sit around this core.

Layer	Examples	Responsibility
Presentation	MainActivity, NewMatchActivity, LiveMatchActivity, ScorecardActivity, GraphsActivity, SettingsActivity	
Scoring / control	MatchManager	Delivery rules, extras, wickets, undo, innings and result logic
Models	Match, Ball, Over, Player, Bowler, Partnership, StandMatch	Standard cricket state
Persistence	MatchHistoryStore, SharedPreferences	History and settings
Output	ScoreXPdfExportUtil / PDF flow	Formatted scorecard document

Application Modules

MODULE	ROLE
Main	Entry point and navigation.
New Match	Team names, overs, toss winner and decision.
Advanced Settings	Players-per-team and special scoring configuration.
Opening Players	Striker, non-striker and opening bowler.
Live Match	Central delivery scoring and match controls.
Scorecard	Detailed innings presentation and PDF access.
Graphs	Runs-per-over, run-rate and innings comparison.
Match History	Local saved matches and deletion.
Match Finished	Completed-match/result flow.
Settings	Preferences, history, PDFs, About and link.
About	Application/project information.

Typical User Journey

1	2	3	4	5	6	7	8	9
Create Match	Configure	Opening Players	Live Scoring	Innings Complete	Second Innings	Result	Scorecard / Graphs	PDF / History

05. Core Data Model & Scoring Workflow

Core Model Objects

Object	Purpose
Match	Match-level score, wickets, balls, innings and partnership state.
Ball	One delivery: runs, extras, legal-ball and wicket information.
Player	Batting statistics and player status.
Bowler	Bowling statistics and special-delivery values.
Over	Over number and delivery/run information.
Partnership	Current/completed partnership data.
MatchManager	Applies rules and coordinates state updates.
SavedMatch	Serializable representation used by Match History.

Delivery Lifecycle

1	2	3	4	5	6	7	8	9
Input	Validate	Apply Runs / Extras	Apply Wicket	Update Player/B owler	Update Partners hip	Update Over	Update Innings	Refresh UI

Live Scoring Workflow

- Create and configure the match.
- Select opening striker, non-striker and bowler.
- Record one delivery at a time.
- Update batsman, bowler, partnership and over information.
- Process special deliveries and wickets.
- Use Undo when the latest delivery is incorrect.
- Complete the first innings and initialize the chase.
- Finish the second innings and calculate the result.
- Review Scorecard, Graphs, History and PDF.

Why Delivery-Level State Matters

A cricket score is not enough by itself. For example, a No Ball + 4 requires the team total, batsman runs, extra total and ball legality to remain consistent. A Run Out + 2 similarly requires both the wicket and completed runs. ScoreX's model-oriented approach supports this multi-value update.

SCORING RULES

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06. Detailed Cricket Scoring Rules

Normal Delivery

A normal scoring action records batsman runs and contributes to the legal-ball count according to the delivery state.

Wide

Wide is an extra. The project contains configurable wide behavior, including the configured run value and re-ball/legal-ball behavior.

No Ball

No Ball records the no-ball extra separately from batsman runs. The project contains configurable no-ball behavior.

Bye

Bye adds to team extras without adding those runs to the batsman's personal run total.

Leg Bye

Leg Bye adds to the appropriate team extras category without becoming batsman runs.

Wicket

A wicket is recorded on the delivery and related wicket/dismissal information is updated.

Run Out

Run Out can include completed runs. The completed runs are retained rather than discarded when the wicket is recorded.

Undo

Undo reverses the latest delivery and the related scoring state.

Over

Legal deliveries advance the over state according to the scoring rules; special deliveries can affect the legal-ball count.

Innings

The first and second innings are maintained separately so the chase and final result can be evaluated.

Run Out Examples

Delivery	Meaning in over	Expected scoring concept
W	Wicket only	Wicket recorded; no completed run.
1 + W	Run Out after one completed run	One completed run + wicket.
2 + W	Run Out after two completed runs	Two completed runs + wicket.
4 + W	Run Out after boundary/completed four	Four completed runs + wicket.

No Ball + 4

The important distinction is that the no-ball extra and batsman runs are separate concepts. The delivery can therefore contribute to the team total through both components while following the project's legal-ball/re-ball

behavior.

SCORECARD & ANALYTICS

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07. Scorecard, Graphs & PDF System

Scorecard

The Scorecard is designed as the main review screen after or during an innings. Its information is organized around cricket concepts rather than a generic data table.

Section	Information
Batting	Player, runs, balls, boundaries and strike rate.
Extras	Total extras and categories.
Bowling	Overs, maidens, runs, wickets and economy-related values.
Fall of Wickets	Wicket sequence with score/over context.
Partnerships / Overs	Supporting innings and over review.

Graphs

GraphsActivity provides visual analysis rather than another table. The project includes runs-per-over, run-rate and innings-comparison concepts. The innings comparison is intended to show cumulative scoring for both innings, with wicket events marked at the relevant scoring position.

Graph Design Goals

- Keep over labels readable and avoid overlapping text.
- Use cumulative score as the main comparison line.
- Place wicket markers on the actual score line rather than floating away from it.
- Make both innings visually comparable even when their lengths differ.
- Keep the graph useful for a quick cricket-style match review.

PDF Scorecard

The PDF output converts scorecard information into a permanent/shareable document. A good PDF should use consistent typography, spacing, section hierarchy, page breaks and table widths so that long names and multiple innings do not collide.

DATA & CONSTRAINTS

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08. Data Storage, Privacy & Constraints

Storage Architecture

The current ScoreX project does not use SQLite, Room, Firebase, MySQL or another traditional database layer. Instead, it uses local Android/Java storage mechanisms.

Data	Mechanism	Purpose
Match History	Java serialization (.ser)	Save/load/delete SavedMatch objects.
Settings	SharedPreferences	Store application/scoring preferences.
PDF exports	Local files / scorex_exports	Keep generated scorecards for sharing/deletion.

Privacy

- Ordinary scoring information remains on the device.
- No cloud synchronization is required for normal scoring.
- Sharing a PDF uses Android's sharing mechanism.
- Clearing application data can remove local history/settings/PDF data.

Constraints

- No remote database or account system in the supplied project.
- No built-in multi-device collaboration.
- Local storage capacity limits retained history/PDFs.
- The application is currently documented as Version 1.0.

SOFTWARE REQUIREMENTS

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09. Software Requirements Specification

ID	Requirement
FR-01	Match creation: accept team names and overs.
FR-02	Toss: store winner and decision.
FR-03	Advanced settings: players-per-team and wide/no-ball behavior.
FR-04	Opening setup: striker, non-striker and bowler.
FR-05	Normal runs: update team/player/bowler state.
FR-06	Extras: Wide, No Ball, Bye, Leg Bye.
FR-07	Wickets: record dismissal information.
FR-08	Run Out: preserve completed runs.
FR-09	Strike: maintain delivery-based strike state.
FR-10	Partnership: maintain current/completed partnerships.
FR-11	Bowling: maintain bowler statistics.
FR-12	Undo: reverse latest delivery.
FR-13	Innings: support first/second innings.
FR-14	Result: calculate/display completion result.
FR-15	Scorecard: display detailed innings information.
FR-16	Graphs: display scoring visualizations.
FR-17	History: save/load/delete locally.
FR-18	PDF: generate/manage/share scorecards.
FR-19	Settings: provide application/scoring controls.
FR-20	Link: provide configured static Google link.

Non-Functional Requirements

- Usability: readable on phone-sized displays.
- Performance: ordinary scoring is local and should not require network access.
- Reliability: Undo and persistence should restore/retain correct state.
- Maintainability: scoring state is organized in dedicated classes.
- Portability: Android API 24+.

User Classes

User	Typical activity
Scorer	Create matches and enter deliveries.

User	Typical activity
Reviewer	Open history, scorecard, graphs and PDFs.
Developer	Maintain scoring rules, UI and persistence.

10. Testing Strategy & Test Cases

Testing Strategy

Testing is centered on scoring correctness, state consistency, UI readability and persistence. Special attention should be given to deliveries that have more than one effect.

Detailed Test Cases

ID	Test	Action	Expected	Priority
T01	Valid match setup	Teams/overs/toss	Opening setup reached	Critical
T02	Invalid overs	Invalid value	Validation/rejection	High
T03	Normal runs	0/1/2/3/4/6	Correct totals	Critical
T04	Wide	Wide delivery	Correct extra/legal behavior	Critical
T05	No Ball	No Ball	Correct extra/legal behavior	Critical
T06	No Ball + 4	No Ball + 4	Extra + batsman runs	Critical
T07	Wicket	Wicket	Wicket state updated	Critical
T08	Run Out	W	Wicket only	Critical
T09	Run Out + 1	1+W	Run retained + wicket	Critical
T10	Run Out + 2	2+W	Run retained + wicket	Critical
T11	Run Out + 4	4+W	Run retained + wicket	Critical
T12	Undo	Undo last run	Previous state	Critical
T13	Undo wicket	Undo wicket	Wicket/state restored	Critical
T14	Undo special	Undo no-ball	Extra/state restored	Critical
T15	Bowling	Multiple bowlers	Statistics consistent	High
T16	Scorecard	Open scorecard	Sections visible	High
T17	Graphs	Multiple overs	Clean rendering	High
T18	Wicket markers	Both innings	Dots on score line	High
T19	First innings	Complete	Second innings ready	Critical
T20	Chase	Reach target	Correct winner	Critical
T21	Defence	Below target	Correct result	Critical
T22	History	Save/load	Match persists	High
T23	History delete	Delete	Removed	High
T24	PDF	Generate	Readable/no overlap	Critical
T25	PDF share	Share	Android share flow	Medium
T26	Settings	Google link	Google opens	Low

Regression Checklist

- MatchManager: rerun runs, extras, wickets, Run Out, no-ball and Undo.
- Scorecard: verify batting, all bowlers, extras, FOW, partnerships and PDF.
- Graphs: verify both innings, labels, cumulative lines and wicket dots.
- PDF: verify long names, multiple rows, page breaks and two innings.
- History: verify save, open and delete.
- Settings: verify preferences and static Google link.

Evidence Note

For a final academic submission, actual screenshots from the tested device should be inserted. This master document does not fabricate screenshots that were not supplied as evidence.

USER MANUAL

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11. User Manual

Quick Start

Create Match → Teams/Overs → Toss → Opening Players → Live Scoring → Innings Completion → Scorecard/Graphs → PDF/History

1. Create a New Match

Step	Action
1	Enter Team A name
2	Enter Team B name
3	Enter number of overs
4	Select toss winner
5	Select Bat or Bowl
6	Start match

2. Opening Players

Select striker, non-striker and opening bowler before the first delivery.

3. Live Scoring

- Use normal run buttons for legal deliveries.
- Use Wide, No Ball, Bye and Leg Bye for extras.
- Use Wicket for dismissals.
- For Run Out with completed runs, include the completed runs in the wicket event.
- Use Undo immediately after an incorrect delivery.
- Open Scorecard or Graphs for review.

4. Scorecard

- Batting — runs, balls, boundaries and strike rate.
- Extras — total and categories.
- Bowling — overs, maidens, runs, wickets and related values.
- Fall of Wickets — wicket sequence.
- Partnerships/overs — supporting innings information.

5. Graphs

Runs-per-over shows scoring volume. Run-rate shows scoring speed. Innings Comparison shows cumulative progress for both innings and places wicket markers at their score positions.

6. Match History

- Open saved matches.

- Review a saved match.
- Delete unwanted local records.

7. PDF

- Open Scorecard.
- Generate PDF.
- Open Saved PDFs.
- Share or delete the document.

8. Settings

- Scorecard style
- Clear Match History
- Saved PDFs
- About ScoreX
- Static Google link
- Scoring/application preferences

9. Good Scoring Practice

- Enter one delivery at a time.
- Verify the score after special deliveries.
- Use Undo instead of manually compensating for mistakes.
- Review the complete scorecard before sharing the PDF.

Q. What is future scope?

A. Cloud backup, online sharing, player/team profiles, tournaments and richer analytics.

Q. Application ID?

A. com.apocalypes.scorex.

Q. Version?

A. 1.0.

CONCLUSION & FUTURE SCOPE

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13. Limitations, Future Scope & Conclusion

Current Limitations

- No cloud synchronization.
- No remote/relational database.
- No user-account system.
- No built-in multi-device collaboration.
- Local storage capacity limits retained records and PDFs.

Future Scope

- Cloud backup and restore.
- Online score sharing through a hosted match link.
- Player and team profiles with career statistics.
- Tournament and points-table management.
- Multi-device scorer collaboration.
- Advanced phase-by-phase analytics.
- Richer bowling spell and batting analysis.
- Automatic PDF backup.

Developer Contributions

Developer	Contribution
Bhatt Parthil	Android application development, scoring workflow, UI integration and project completion.
Nimavat Viraj	Development support, testing and project documentation.

Conclusion

ScoreX provides an end-to-end local Android cricket scoring workflow. Its main strength is the coordinated handling of delivery events and the resulting player, bowler, partnership, innings and match information. The project extends beyond simple score entry with scorecards, visual analytics, local history and PDF output.

Documentation Integrity

This master document intentionally stays within the capabilities represented by the supplied ScoreX project. It does not claim a database, cloud service or other feature that is not supported by the source.